

Assignment 1: Intro to World Models and Control

COMP 765 - Fall 2026

Question 1 - Profile of a World Model: DreamerV3

I chose DreamerV3's recurrent state-space model (RSSM) because it connects prediction with control: the agent learns what could happen after an action and uses those predictions to improve its policy [1].

Inputs and outputs. For training, Dreamer uses sequences of observations, actions, rewards, and flags indicating whether the episode continues. Observations can be RGB images or low-dimensional state vectors. The RSSM combines a memory state h_t with categorical latent variables z_t . The encoder uses the current observation and memory to infer the latent state; the dynamics model predicts it without seeing that observation. From the combined state, the model reconstructs the observation and predicts reward and continuation. The actor selects a discrete or continuous action, while the critic estimates a distribution over future return [1].

Design principles and differences. The part that makes this useful for control is imagination. Dreamer trains its actor and critic on short trajectories predicted in latent space, using predicted rewards and bootstrapped lambda-returns to account for rewards beyond the rollout. This avoids having to generate full images for every policy update. Once trained, the actor selects actions directly, without searching over action sequences at every step. Compared with the hand-derived model used for LQR here, Dreamer learns its dynamics and representation from experience. Unlike MuZero's task-focused model, it also reconstructs observations [1,2].

Theory and key findings. One useful piece of the theory is the KL loss between the observation-based latent distribution and the model's prediction. Separate stop-gradient losses train the predictor to match the encoded state and encourage that state to be predictable. A one-nat free-bits threshold stops this penalty from dominating when the distributions already agree well. V3 also mixes in 1% uniform probability, uses symlog to compress large signed values, predicts rewards and returns with two-hot distributions, and normalizes returns using percentiles. Together, these changes make training less sensitive to the task's scale. The authors report results on over 150 tasks with fixed hyperparameters, including Control Suite, Atari, and Minecraft [1].

Foundational papers and my assessment. The earlier papers help explain how the model developed. DreamerV1 used continuous stochastic states and propagated gradients through imagined trajectories to learn behaviour [2]. DreamerV2 switched to categorical states with straight-through gradients and reached human-level Atari performance [3]. V3 keeps the discrete RSSM but focuses on making one configuration work across domains [1]. The public JAX implementation includes training code and configurations [4]. My main takeaway is that V3 makes world-model learning more reliable across tasks, but prediction errors can still mislead the policy and training needs substantial computation. I would want to check how well its predictions hold up in states outside its training experience.

Sources for Q1: [1] Hafner et al., Mastering Diverse Domains through World Models, <https://arxiv.org/abs/2301.04104>. [2] Hafner et al., Dream to Control, <https://arxiv.org/abs/1912.01603>. [3] Hafner et al., Mastering Atari with Discrete World Models, <https://arxiv.org/abs/2010.02193>. [4] Official implementation, <https://github.com/danijar/dreamerv3>.

Question 2 - Model and Control the Cart-Pole

A. Linearization about the upright equilibrium

The simulator state is $s = [x, \dot{x}, \theta, \dot{\theta}]^T$. Define the local angle $\phi = \theta - \pi$ and deviation state $\delta = [x, \dot{x}, \theta, \dot{\theta}, \phi]^T$. Near $\theta = \pi$, $\sin(\theta)$ is approximately $-\phi$, $\cos(\theta)$ is approximately -1 , $\cos(\theta)^2$ is approximately 1 , and products such as $\dot{\theta}^2 \sin(\theta)$ are higher order and discarded. Let $D = 4(M+m) - 3m$.

$$\begin{aligned} \ddot{x} &= [3m g \phi - 4b \dot{x} + 4u] / D \\ \ddot{\theta} &= [6(M+m)g \phi - 6b \dot{x} + 6u] / (lD) \end{aligned}$$

Therefore $\delta\dot{ = A \delta + B u$, with the state order used by the code:

A =	[0 1 0 0]	B =	[0]
	[0 -4b/D 0 3mg/D]		[4/D]
	[0 -6b/(lD) 0 6(M+m)g/(lD)]		[6/(lD)]
	[0 0 1 0]		[0]

With $M=m=l=0.5$, $g=9.82$, $D=2.5$. The handout states $b=0.1$, giving cart and pole damping entries of -0.16 and -0.48 . The executable CartPole class uses $b=1.0$, so the implemented model uses the following values; all other entries agree:

A =	[0 1.0 0 0]	B =	[0]
	[0 -1.6 0 5.892]		[1.6]
	[0 -4.8 0 47.136]		[4.8]
	[0 0.0 1 0]		[0]

B. Balancing: LQR design and stability

I used $Q = \text{diag}(2, 1, 2, 120)$ and $R = [0.2]$. The largest weight is on angle error to prioritize balancing; cart-position and velocity penalties encourage recentering and damping, while R penalizes force. The continuous algebraic Riccati equation returns $K = [-3.1623, -6.4776, 7.2892, 47.0906]$. The code applies $u = K(g-s)$, with the angular component wrapped to $[-\pi, \pi]$. Equivalently, $u = -K \delta$. The closed-loop poles are -16.7176 , -7.7261 , and $-0.8903 \pm j 0.6012i$, so the linear model is asymptotically stable. For comparability with swing-up, force is limited to ± 30 N.

B(i). Balancing and initial-offset tests

Each deterministic run used the nonlinear supplied ODE simulator for 20 s at $dt=0.005$ s. Success means that throughout the final 5 s, $|\theta-\pi|<0.10$ rad and $|x|<0.50$ m. Capture means $|\theta-\pi|<0.10$ rad and $|\dot{\theta}|<0.25$ rad/s. The same Q, R, K, and force limit were used for every trial.

initial offset	capture time (s)	max $ u $ (N)	final-window max $ \text{angle} $	result
default ($\pi/40$)	0.000	3.70	6.34e-08	PASS
0.01	0.000	0.47	8.04e-09	PASS
0.10	0.000	4.71	8.09e-08	PASS
$\pi/8$	0.380	18.49	3.51e-07	PASS
$\pi/4$	1.205	30.00	9.88e-07	PASS

The default start and all four requested offsets balance. The $\pi/4$ case hits the 30 N force limit and takes 1.205 s to meet the capture criterion; smaller tested errors remain unsaturated. The largest successful tested offset is $\pi/4$ (0.785 rad), with all other initial state deviations zero. These sampled results do not certify a continuous range or global stability: the linearization and Riccati proof are local, and saturation alters the nonlinear closed loop.

C. Swing-up from the downward configuration

Initial performance (i). With initial-offset= π , the simulator begins at $\theta=2\pi$, physically downward. Pure LQR performs poorly: the local angle error is maximally ambiguous at $\pm\pi$, the command saturates, and the pole does not settle upright during 20 s. Its final-window maximum angle error is 3.1411 rad. Tuning Q and R changes local aggressiveness but does not give LQR a global energy-building strategy.

Changes explored (ii). I therefore explored a hybrid controller. Away from upright it shapes the pole energy $E = (m l^2/6) \dot{\theta}^2 + (mgl/2)(1-\cos(\theta))$, with target $E^* = mgl$, using $u = -k_E(E^*-E) \text{sign}(\dot{\theta} \cos(\theta)) - k_x x - k_v \dot{x}$, clipped to ± 30 N. I used $k_E=40$, $k_x=1$, and $k_v=2$. When $|\theta-\pi|<0.42$ rad and $|\dot{\theta}|<3.5$ rad/s, control switches to LQR; 0.65 rad hysteresis prevents chatter.

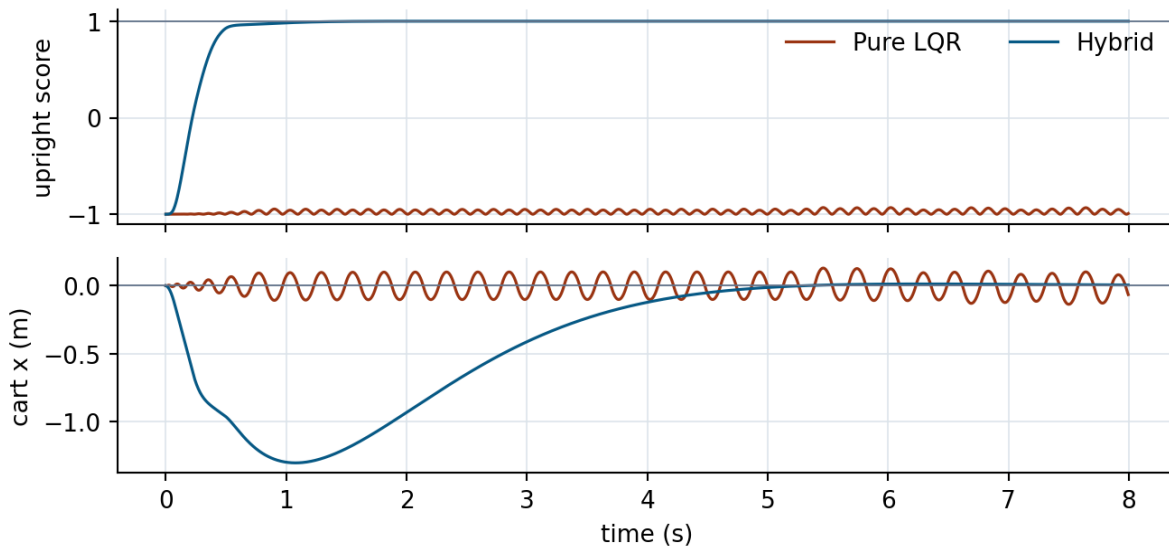


Figure 1. Downward-start comparison. Upright score $\cos(\theta-\pi)$ is +1 upright and -1 downward. Energy shaping swings up the pole; LQR then recenters the cart.

C(iii). Best single-pole swing-up performance

The hybrid controller first meets $|\theta - \pi| < 0.10$ rad and $|\dot{\theta}| < 0.25$ rad/s at 1.340 s. It passes the 20 s success criterion: over the final 5 s, maximum angle error is $1.14\text{e-}06$ rad and maximum $|x|$ is $4.33\text{e-}06$ m. Peak force is 30 N. This is my best swing-up result. It is stronger than pure LQR but should be interpreted as a deterministic simulation result; actuator delay, sensor noise, and model mismatch were not tested.

D. DoubleCartpole bonus

For the six-state system $[x, \dot{x}, \theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2]$, let $\phi_1 = \theta_1 - \pi$ and $\phi_2 = \theta_2 - \pi$. At the upright equilibrium, the simulator's coupled acceleration equations simplify to $H_0 \ddot{a} = r$, with $a = [x, \ddot{x}, \ddot{\theta}_1, \ddot{\theta}_2]^T$:

$$H_0 = \begin{bmatrix} [3, 0.9, 0.3], & [4.5, 2.4, 0.9], & [3, 1.8, 1.2] \end{bmatrix}$$

$$r = [-0.2 \ddot{x} + 2u, 44.19 \phi_1, 29.46 \phi_2]^T$$

Substituting $a = H_0^{-1}r$ into the kinematic equations gives $\dot{\delta} = A \delta + Bu$. With $Q = \text{diag}(2, 1, 2, 2, 120, 120)$, $R = [0.2]$, the continuous LQR gain is $K = [3.1623, 6.8993, 0.8285, 27.1835, -135.5356, 189.0925]$. All six eigenvalues of $A-BK$ have negative real part. A central-difference check against the nonlinear simulator agrees with the derived A and B to about $1\text{e-}8$.

start offset	capture (s)	tail max angle (rad)	peak $ x $ (m)	result
default	0.00	$1.05\text{e-}06$	0.273	PASS
0.01	0.00	$1.34\text{e-}07$	0.035	PASS
0.10	0.49	$1.33\text{e-}06$	0.347	PASS
$\pi/8$	1.48	$4.71\text{e-}06$	1.23	PASS
0.42	--	3.14	242	FAIL
$\pi/4$	--	3.14	245	FAIL
π	--	3.14	334	FAIL

Both poles balance from the default start through $\pi/8$ with a 40 N cap. The $\pi/8$ trial briefly moves the cart 1.23 m, then recenters it; 0.42 rad, $\pi/4$, and the downward start do not recover. A 30 N cap also failed at $\pi/8$. The largest successful tested offset is 0.393 rad, and the next tested offset (0.42 rad) fails; no untested initial conditions are claimed.

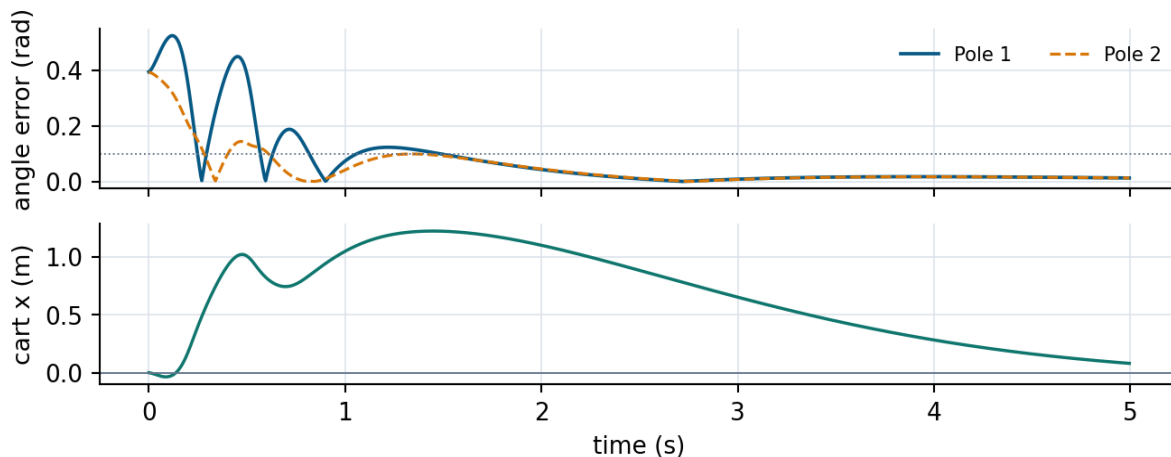


Figure 2. Both pole angles and cart position during the $\pi/8$ recovery.

E. Research bonus - learned world model

I collected 7,680 training and 1,920 held-out transitions with randomized initial states and forces, storing each row in the simulator's (next state, state, force) format. A compact action-conditioned model fits trigonometric features of the two accelerations by least squares. Each output has the learned form $\text{acceleration} = \text{feature numerator} / (1 - a \cos^2(\theta))$; fourth-order integration predicts the next state. The fit uses transition data and no simulator constants.

On 12 held-out 0.8 s rollouts driven by recorded forces, root mean square error is $4.00\text{e-}05$ m in cart position and $1.20\text{e-}04$ rad in pole angle. A 50 ms receding-horizon controller evaluates candidate forces with the learned model, a quadratic running cost, and an LQR terminal value. It balances offsets 0.1, $\pi/8$, and $\pi/4$ in separate 8 s simulator trials; final 2 s angle errors stay below 0.0037 rad. This demonstrates model use in control, though the planner is not shown to outperform LQR or swing up from downward.

Held-out prediction and control

state variable	one-step RMSE	0.8 s open-loop RMSE
cart position (m)	4.47e-08	4.00e-05
pole angle (rad)	1.83e-07	1.20e-04

start offset	tail max angle (rad)	tail max x (m)	peak u (N)
0.1	0.00024	0.00128	4.0
$\pi/8$	0.00102	0.00547	12.0
$\pi/4$	0.00361	0.0177	18.0

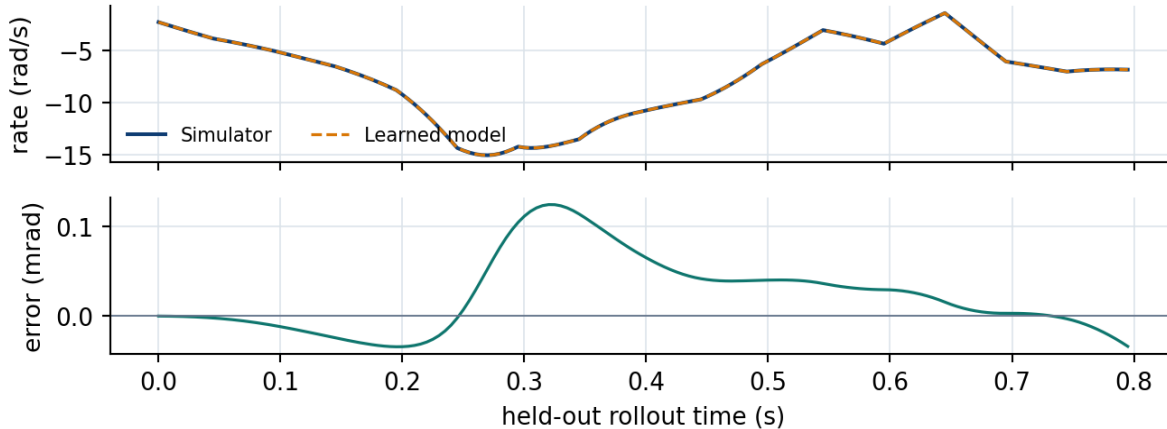


Figure 3. One unseen trajectory: predicted pole rate follows the simulator; lower panel shows accumulated angle prediction error in milliradians.

Reproduce all results from assignment1/TeachingCartpole with python3 experiment.py, python3 double_experiment.py, and python3 world_model_bonus.py. The simulator source is D. Meger et al., <https://github.com/dmeger/TeachingCartpole>.